

Independent external validation of the SugarCode AI computational-biology platform against held-out public benchmarks

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Abstract

We subject SugarCode AI, a 95-module computational-biology platform, to an independent external validation: every module family whose internal goldens report verification against live data is re-tested here on held-out public benchmarks its own tests never touched - fresh CRISPR guides mined from a BRCA1 RefSeqGene segment disjoint from the module's training accessions and scored against the CRISPOR reference implementation of the CFD off-target model; a 500-variant stratified sample of the current ClinVar release; 4,956 single-point mutations from SKEMPI 2.0; the BiGG *E. coli* core metabolic model; and a globin-family alignment produced by EBI Clustal Omega. We report per-module agreement metrics with bootstrap confidence intervals, publish all fixtures with provenance, and document failure modes without suppressing negative results.

1 Introduction

Computational-biology toolkits are typically validated against the same data that informed their development, inflating confidence. Independent, held-out validation against public reference implementations and current database releases is the standard this platform must meet to be usable in laboratory settings. This study asks a single question: does each verified claim in the platform's own status ledger survive contact with data it has never seen? We answer it module family by module family, with frozen fixtures, pinned random seeds, and metrics reported with 95% bootstrap confidence intervals. All code, fixtures, and results are released in this repository.

2 Related Work and Benchmarks

Pending validation runs; filled on completion.

3 Methods

For each module family we constructed a held-out test disjoint from every fixture used during platform development. Negative results are preserved and reported alongside positives; no test was re-designed after seeing its outcome.

3.1 Data and Provenance

All datasets are public and carry recorded provenance (see `data/README.md`): NCBI ClinVar `variant_summary.txt.gz` (retrieved 2026-09-24; 500-variant stratified reservoir sample from 1,517,551 eligible GRCh38 SNVs with criteria-provided review status); SKEMPI 2.0 [?] (150-mutation frozen subset, seed 7); BiGG `e_coli_core` [?]; BRCA1 RefSeqGene NG_005905.2:60000-70000 (NCBI efetch); CRISPOR reference distribution [?]; 25 reviewed globin-family UniProtKB sequences aligned with EBI Clustal Omega [?].

3.2 Mathematical Formulation

CFD off-target scoring follows Doench et al. 2016 [?]: $CFD = \prod_i m_i \cdot p(\text{PAM})$, with per-position mismatch penalties m_i from the reference distribution. Binding free-energy changes are $\Delta\Delta G = RT \ln(K_d^{mut}/K_d^{wt})$ at $T = 300$ K ($RT = 0.596$ kcal/mol). Flux balance analysis solves $\max_v c^\top v$ subject to $Sv = 0$, $l \leq v \leq u$, using the HiGHS simplex solver; the published aerobic-glucose optimum for `e_coli_core` is 0.8739 h^{-1} [?]. Agreement is quantified with Pearson r , Mann-Whitney AUC, and binary-classification metrics; confidence intervals are percentile bootstrap with 1,000 resamples at fixed seed.

3.3 Implementation and Testing

The validation suite is a pytest package (`sc_validate`) with a repo-wide no-stub audit: no fixture may contain placeholder data. Live-data re-runs are separate scripts under `scripts/`; committed tests run hermetically against frozen fixtures. CI runs the suite on every push.

4 Results

Pending validation runs; filled on completion.

5 Benchmark Comparison

Pending validation runs; filled on completion.

6 Failure Modes and Negative Results

Pending validation runs; filled on completion.

7 Discussion

Pending validation runs; filled on completion.

8 Conclusion

Pending validation runs; filled on completion.

References

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